Comparative Phylogenetic Analysis of Mnd1, Hop2, and Dmc1 Across Metamonada

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Author Contributions: AE: Analysis, Investigation, Visualization, Writing; CN: Analysis, Investigation, Visualization, Writing; PR: Analysis, Investigation, Visualization, Writing

Competing Interest Statement: The author(s) declared no competing interest.

Classification: Biological Sciences: Evolution

Keywords: Metamonada, Protists, Phylogenetics, Homologous Recombination, Meiosis

Abstract

The superphylum Metamonada represents a broadly diverse group of anaerobic protists that have shown interesting evolutionary developments, including the complete loss of mitochondria in some species. To explore potential factors in this group's genetic evolution, this paper looks at the gene phylogeny of three different proteins significant to homologous recombination in meiosis: Hop2, Mnd1, and Dmc1. The Dmc1-mediated DNA repair pathway, regulated by a Hop2-Mnd1 heterodimer, provides an interhomolog alternative to Rad51, which is more prevalent across eukaryotes and uses an intersister repair mechanism for dsDNA breaks during recombination. Further knowledge of this interhomolog mechanism could shed light on factors behind quicker evolutionary developments and vast speciation across Metamonada. We hypothesized that Dmc1, Hop2, and Mnd1 evolved together and may have followed a pattern of vertical transmission following Dmc1 gene duplication events within Metamonada in response to the loss of Rad51A. In this study, we used the EukPhylo pipeline to align sequences from the proteins of interest retrieved from protist transcriptomes and built phylogenetic trees to analyze for significant evolutionary patterns. Our findings indicate that while Hop2 and Mnd1 show similar phylogenies, there is insufficient evidence to conclusively determine whether they may have co-evolved. Dmc1 exhibited a more complex phylogenetic distribution across Metamonada, with multiple homologs recovered across several metamonad lineages suggesting potential duplication events. This phylogenetic analysis partially characterizes the evolution of the Dmc1-mediated DNA repair pathway in metamonads and provides insights into the development of meiotic mechanisms that contribute to genomic evolution within protists.

Introduction

Metamonada is a superphylum of anaerobic protists that represents an early-diverging lineage of eukaryotes (Al Jewari et al., 2023; Williamson et al., 2025). Metamonads have reduced mitochondria to varying degrees, with some species even having lost them completely. Although metamonads are not the only eukaryotic group with reduced mitochondria, the discovery of the first completely amitochondriate eukaryote within the Oxymonadida order of the Preaxostyla phylum marks a step further in the evolutionary loss of this organelle (Novák et al., 2023; Williams et al., 2024). Metamonada is composed of four major phyla (Parabasalia, Fornicata, Preaxostyla, and Anaeramoebae) along with a currently unclassified clade referred to as BaSk, and ranges from relatively morphologically simple to large and complex and from symbiotic to free-living. Even just within Parabasalia, recent research indicates that the sub-group of free-living trichomonads shows significant diversity in both morphological characteristics and ecological niche (Céza et al., 2022). Due to their broad range of diversity and this uniquely significant mitochondrial loss, Metamonada's evolution has been a subject of interest in microbiological research.

As a mechanism of gene conversion within eukaryotes, homologous recombination represents a significant factor in genome evolution and could consequently play a role in the rate at which the Metamonada lineages shift over time. Major proteins within this process include Radiation Sensitive 51 (Rad51) and Disruptive Meiotic cDNA 1 (Dmc1), two recombinases that work within DNA repair pathways to establish nucleoprotein filaments and locate target sequences to pair, as well as the upstream Homologous-pairing Protein 2 (Hop2) and Meiotic Nuclear Division Protein 1 (Mnd1) proteins that form a heterodimer regulating their activity (Tsubouchi et al., 2002; Petukhova et al., 2005; Pezza et al., 2014). While Rad51 is not dependent on the presence of the Hop2-Mnd1 heterodimer, its intersister DNA repair pathway presents less opportunities for genetic changes and diverse evolution in contrast to the heterodimer-dependent Dmc1-mediated interhomolog repair, which allows for complete gene replacement and further genome shuffling (Uanschou et al., 2013). Despite this, Rad51 has been found in most eukaryotic genomes to date, whereas Hop2, Mnd1, and Dmc1 are restricted to certain lineages, indicating separate evolutionary paths and potentially a later evolutionary development (or gene loss) for Dmc1-mediated repair (Chen et al., 2004).

Dmc1's broader evolutionary history has already been investigated in prior research, including its distribution across eukaryotes and lineage-specific losses (Lin et al., 2006), and apparent duplication of this gene in relation to the loss of a Rad51A protein has been noted within the Fornicata phylum (Salas-Leiva et al., 2021). However, its broader patterns remain unresolved and its evolutionary relationship to Hop2 and Mnd1 is unclear, raising questions about how this alternative pathway to Rad51 that may promote stronger genetic diversity could have arisen within metamonads. The development of this interhomolog DNA repair pathway in homologous recombination within eukaryotes could help researchers further understand the

mechanisms behind which protists develop or lose genes, which in turn could also further studies on the evolution of amitochondriate organisms and how they may continue to evolve over time.

We hypothesized that Dmc1's gene duplication events within Metamonada, as suggested in Salas-Leiva et al. (2021), could have enabled the further proliferation of a Dmc1-mediated pathway in later lineages in place of Rad51. Considering their functional interdependence, Hop2, Mnd1, and Dmc1 may have co-evolved along this DNA repair pathway as a more advantageous alternative to its more prevalent Rad51-mediated counterpart. To assess whether these three proteins may have evolved along similar paths and how they may have individually arisen throughout protists and into Metamonada, we analyzed their phylogenetic relationships and built gene trees to compare to each other and to known species phylogenies. This paper investigates the phylogeny of each gene individually and then discusses the evolutionary implications of their potential origins and how they may have evolved together to promote broader diversity within the Metamonada group.

Materials & Methods

Computational analyses were performed using Arizona State University's Sol high-performance computing (HPC) cluster. Phylogenetic analyses focused on three proteins involved in meiotic homologous recombination: Hop2 (orthogroup OG6_103529), Dmc1 (orthogroup OG6_100483), and Mnd1 (orthogroup OG6_103073). Metamonad transcriptomes assembled using rnaSPAdes (Bushmanova et al., 2019) were processed through Part 1 of the EukPhylo pipeline (Katz et al., 2025), which included contamination assessment and extraction of predicted proteins assigned to orthogroups using the HOOK database. This initial processing was completed before the present analyses, and the resulting processed datasets were used as input for EukPhylo Part 2. Part 2 of EukPhylo was subsequently used to retrieve homologous sequences corresponding to each selected orthogroup from the processed transcriptomic and genomic datasets. A similarity filter with a cutoff of 0.99 was applied to remove highly similar sequences from the same sample while retaining the sequence with higher coverage. The pipeline generated orthogroup-specific sequence alignments and preliminary maximum-likelihood phylogenies, which served as the starting datasets for subsequent analyses.

Following the initial EukPhylo analysis, each orthogroup-specific phylogeny and corresponding sequence alignment was examined for additional redundant sequences. Dataset pruning was performed using Taxus (Seraff, n.d.) when redundancy was identified. The Hop2 dataset was not additionally pruned because no groups of repeated sequences requiring removal were identified. For Dmc1, redundant non-Metamonada sequences were removed when multiple sequences from the same taxon occurred within the same clade, retaining the sequence with the highest coverage. A total of 91 Dmc1 sequences were removed. For Mnd1, repeated sequences were evaluated based on their phylogenetic placement and sequence coverage. Closely clustered sequences that appeared redundant were removed, whereas sequences from the same taxon occurring in different regions of the tree were retained because they could represent distinct homologs. A total of 10 Mnd1 sequences were removed.

Following pruning, the sequence datasets were realigned using MAFFT (Katoh & Standley, 2013) and trimmed using trimAl (Capella-Gutiérrez et al., 2009) with the -automated1 option to select an appropriate trimming strategy for each protein alignment. Maximum-likelihood phylogenies were reconstructed using IQ-TREE 2 (Minh et al., 2020) under the LG+F+G4 amino acid substitution model. Branch support was evaluated using 1,000 ultrafast bootstrap replicates (Hoang et al., 2018).

Final phylogenies were visualized using FigTree (Rambaut, 2018) and prepared for presentation using appropriate figure-editing software. Metamonada lineages were highlighted to facilitate comparison among the three protein phylogenies, while selected non-Metamonada background clades were collapsed to improve readability without obscuring relevant Metamonada placements. The Hop2 phylogeny was rooted between Diphora+ and Opimoda+ taxa, whereas the Dmc1 and Mnd1 phylogenies were left without an assigned biological root. Bootstrap support values were displayed as specified in the corresponding figure legends.

Results

Maximum-likelihood phylogenies were reconstructed for Mnd1, Hop2, and Dmc1 following sequence processing and phylogenetic analysis (Figures 1–3). The initial datasets contained 527 Mnd1, 333 Hop2, and 1,222 Dmc1 sequences. Following redundancy assessment and pruning, the final phylogenies contained 517 Mnd1, 333 Hop2, and 1,131 Dmc1 sequences. Metamonada homologs occurred in multiple regions of all three protein phylogenies rather than forming a single Metamonada clade, resulting in a polyphyletic distribution within the gene trees. The representation and phylogenetic distribution of individual Metamonada groups differed among the three proteins.

The Mnd1 phylogeny contained representatives of all five major Metamonada groups examined: Parabasalia, Fornicata, Preaxostyla, BaSk, and Anaeramoebae (Figure 1A–D). Parabasalia was extensively represented, with homologs occurring in several regions of the phylogeny. Fornicata was also recovered in multiple regions and included representatives such as *Giardia intestinalis*, *Spironucleus salmonicida*, *Carpediemonas membranifera*, and *Trepomonas steinii*. Preaxostyla homologs likewise occurred in multiple regions. BaSk and Anaeramoebae homologs clustered within particular regions of the Mnd1 phylogeny, whereas Parabasalia, Fornicata, and Preaxostyla did not form single monophyletic groups. Several relationships within Metamonada-containing regions received strong ultrafast bootstrap support (UFBoot ≥95), although support was not uniformly high across all relationships among the separated Metamonada placements. Multiple individual taxa contained more than one recovered Mnd1 homolog. These were not interpreted as confirmed gene-duplication events because the Mnd1 phylogeny alone did not establish duplication events.

The Hop2 phylogeny contained 333 sequences and included representatives of Parabasalia, Fornicata, Preaxostyla, and BaSk (Figure 2). Parabasalia was the most extensively represented Metamonada group, including Trichomonadea, Tritrichomonadea, Cristamonadea, Trichonymphea, and Spirotrichonympha. Fornicata formed another prominent region and

included *Giardia intestinalis, Spironucleus salmonicida, Trepomonas, Retortamonas, and Gyromonas ambulans*. Preaxostyla was represented by *Trimastix marina* and *Paratrimastix pyriformis*, while BaSk was represented by *Skoliomonas*. No Anaeramoebae homologs were identified. Metamonada homologs showed a polyphyletic distribution within the Hop2 gene tree. Bootstrap support varied across the phylogeny, with several relationships within major Metamonada-containing regions receiving high support while deeper relationships showed lower support.

The Dmc1 phylogeny contained 1,131 sequences and included all five major Metamonada groups examined (Figure 3A–D). Fornicata homologs did not form a single monophyletic group, occurring in three regions that included a large Fornicata clade and two additional placements containing *Trepomonas* sequences. BaSk homologs occurred in separate regions and included multiple *Skoliomonas* and *Barthelona* sequences. Preaxostyla showed a similar polyphyletic distribution, with Oxymonadida represented in three regions and *Paratrimastix* and *Trimastix* occurring separately. Parabasalia was represented in more than one region, whereas Anaeramoebae homologs clustered within a single region of the tree. Multiple Dmc1 homologs were recovered within Fornicata, BaSk, and Preaxostyla/Oxymonadida. Many relationships within individual Metamonada clades had strong ultrafast bootstrap support (UFBoot ≥95), whereas several deeper relationships connecting separated clades to surrounding eukaryotic sequences had lower support.

Comparison of the three protein phylogenies revealed both shared and protein-specific patterns (Figures 1–3). Parabasalia, Fornicata, and Preaxostyla were represented in all three phylogenies. Mnd1 and Dmc1 contained all five major Metamonada groups examined, whereas Hop2 contained a single BaSk representative and no identified Anaeramoebae homologs. Multiple homologs from individual Metamonada taxa occurred in both the Mnd1 and Dmc1 phylogenies, although this pattern was more widespread in Dmc1. Overall, the three gene trees differed in the representation, distribution, and support of Metamonada homologs despite each showing a polyphyletic pattern.

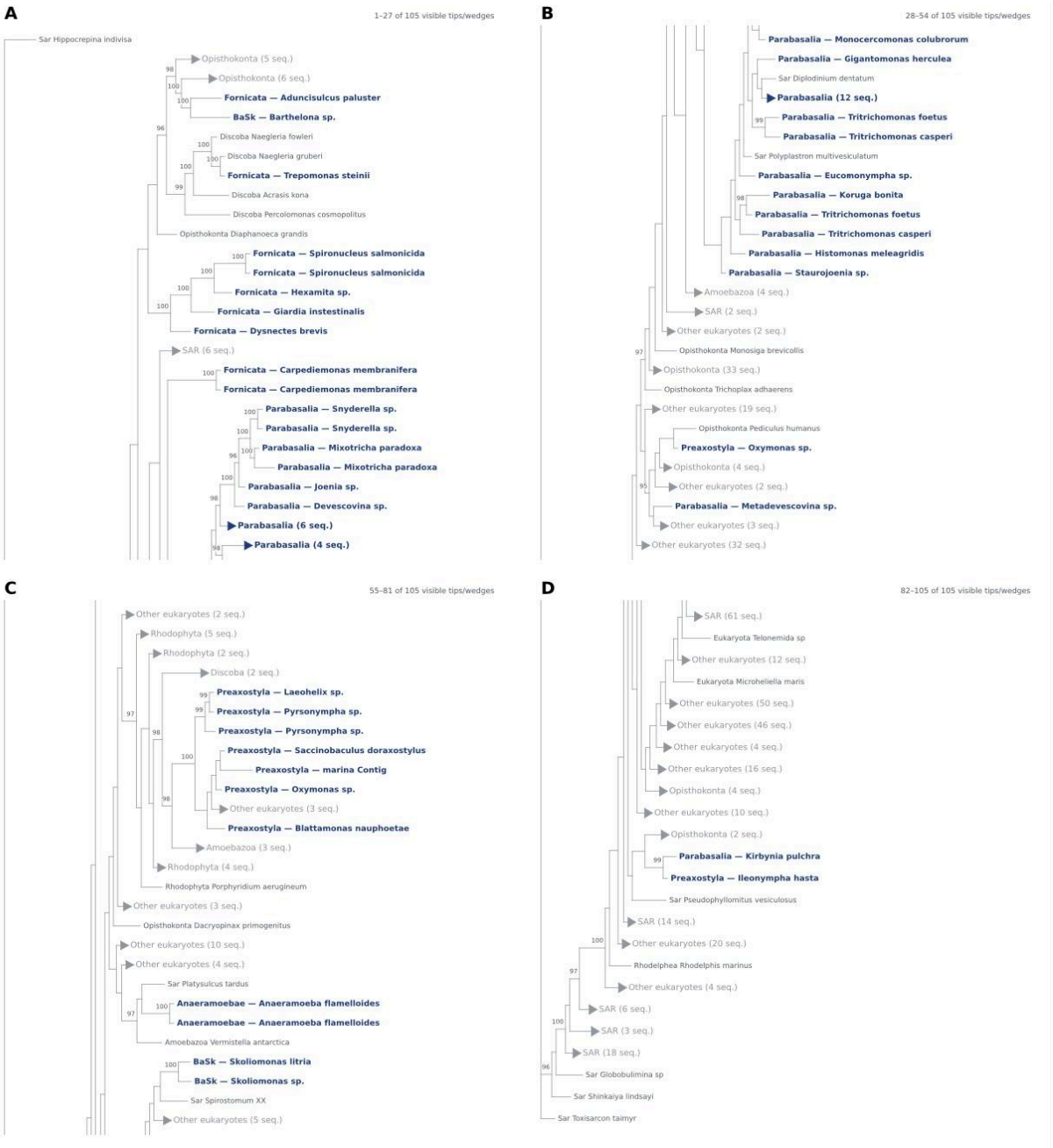


Figure 1. Distribution of Mnd1 homologs across Metamonada and other eukaryotes. *Maximum-likelihood phylogeny of 517 Mnd1 homologs shown as four consecutive portions of the same phylogeny (A-D). Metamonada sequences are shown in blue, while non-Metamonada sequences are shown in grey. Large background clades were collapsed to improve readability and are labeled according to their predominant taxonomic composition and the number of sequences contained within each clade. Metamonada-containing regions were left expanded to show the distribution of Mnd1 homologs across the phylogeny. Ultrafast bootstrap support*

values ≥95 (UFBoot ≥95) are shown at internal nodes. The phylogeny was not assigned a biological root.

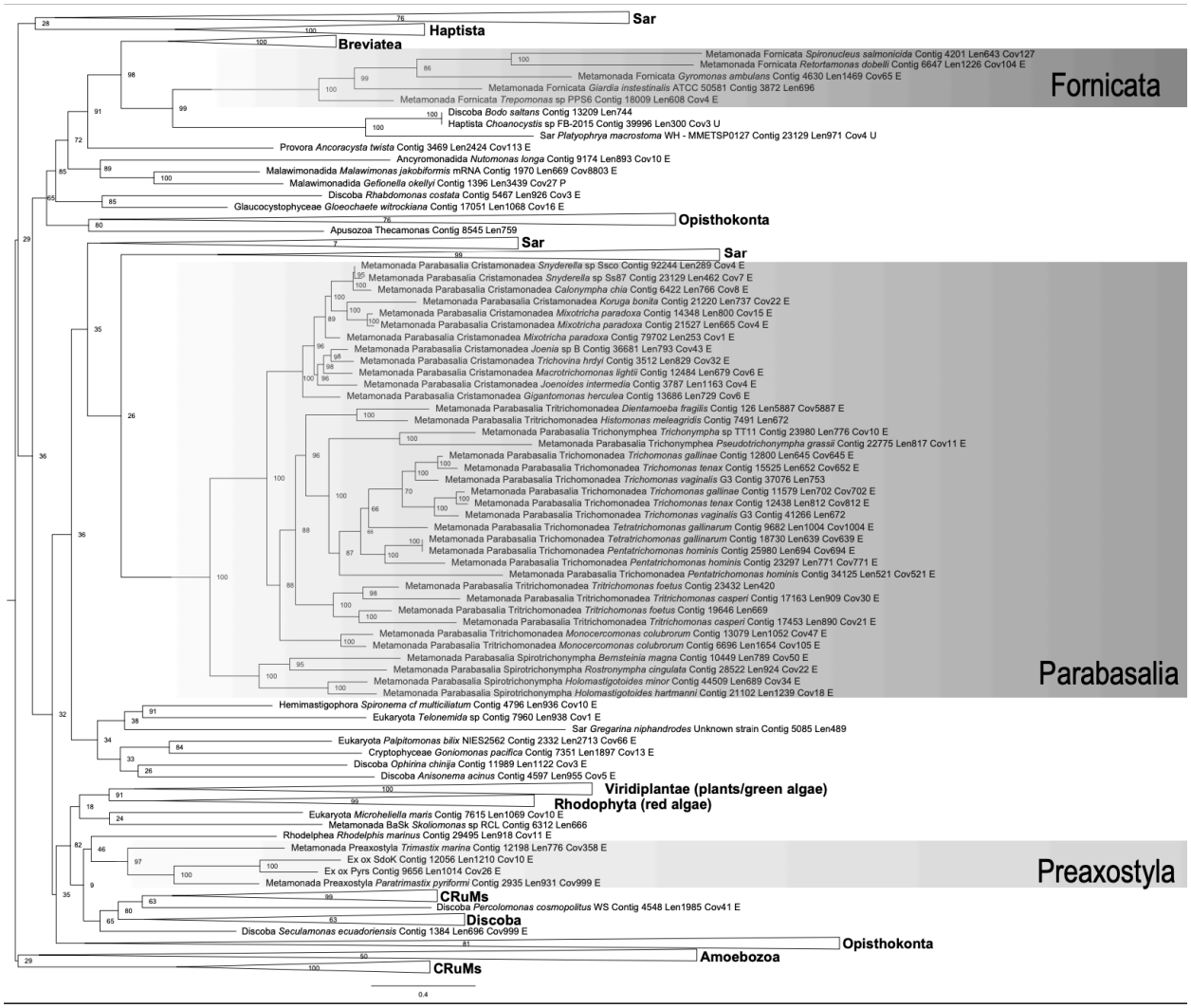


Figure 2. Distribution of Hop2 homologs among major Metamonada lineages.
Maximum-likelihood phylogeny of 333 Hop2 homologs showing the distribution of Metamonada sequences among Fornicata, Parabasalia, Preaxostyla, BaSk, and other eukaryotic groups. Major Metamonada-containing regions are highlighted with transparent grey shading, with Fornicata, Parabasalia, and Preaxostyla labeled directly on the tree. Non-Metamonada clades containing more than two sequences were collapsed and labeled according to their predominant taxonomic composition to improve readability. UFBoot support values are displayed at internal nodes. The phylogeny was rooted between Diphora+ and Opimoda+ taxa. The scale bar represents amino acid substitutions per site.

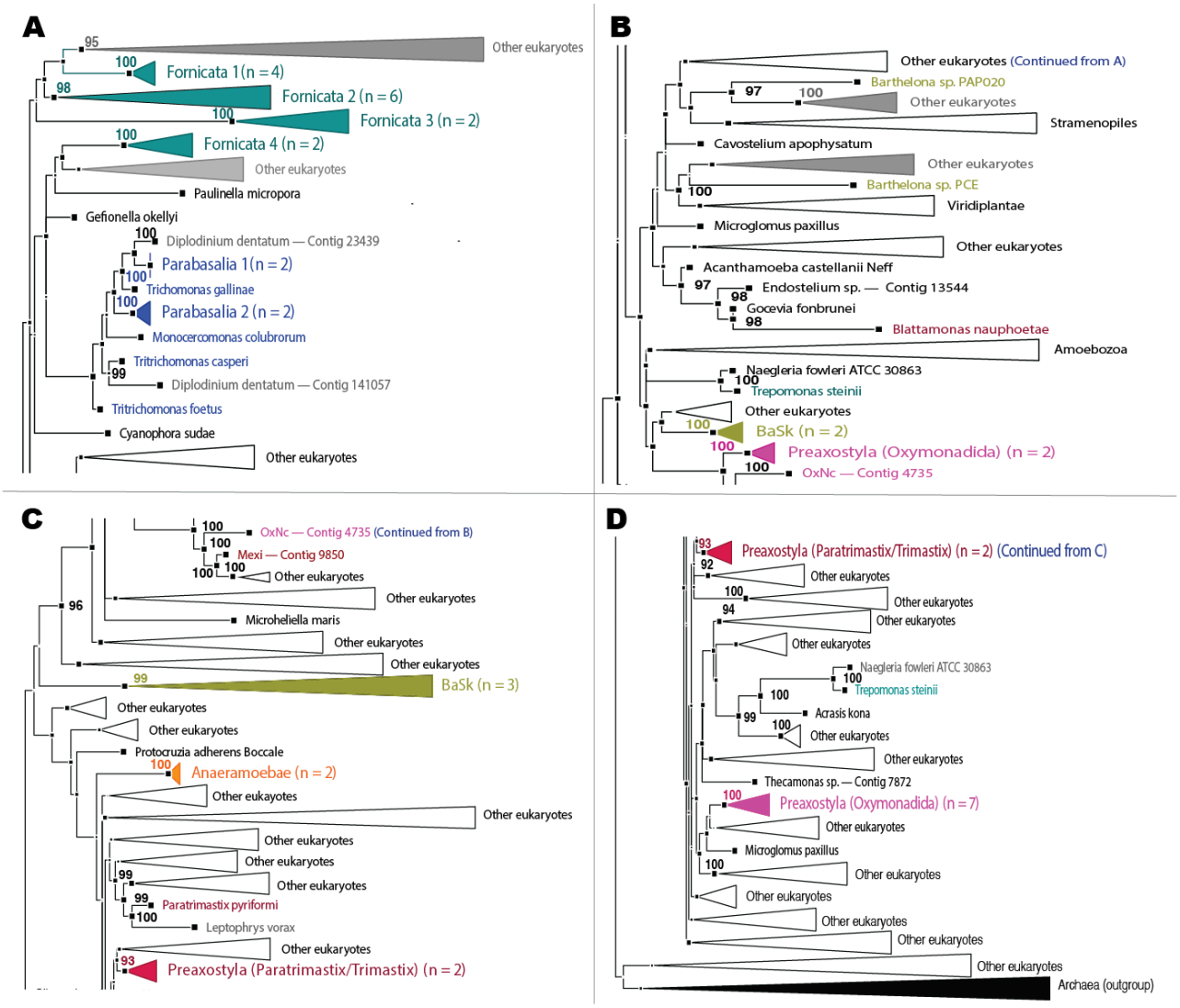


Figure 3. Phylogenetic distribution of Dmc1 homologs across Metamonada.
Maximum-likelihood phylogeny of Dmc1 homologs shown as four consecutive portions of the same phylogeny (A-D). Metamonada lineages are color-coded: Fornicata (teal), Parabasalia (blue), BaSk (olive), Anaeramoebae (orange), Preaxostyla (Paratrimastix/Trimastix; red), and Preaxostyla (Oxymonadida; magenta). Colored triangles represent collapsed Metamonada clades, with n indicating the number of sequences contained within each collapsed clade. Non-Metamonada eukaryotic clades are collapsed in gray, while archaeal sequences are shown in black and positioned at the base of the displayed phylogeny without assigning a biological root. Triangle widths are schematic and do not represent the number of sequences within collapsed clades. Ultrafast bootstrap support values ≥95 (UFBoot ≥95) are shown at internal nodes. Continuation labels indicate connections between consecutive portions of the phylogeny.

Discussion

We originally hypothesized that Hop2, Mnd1, and Dmc1 may have followed similar evolutionary histories because of their functional relationship in homologous recombination (Chen et al., 2004). However, analysis of the three protein phylogenies suggests a more complicated evolutionary history. Mnd1 was broadly distributed across Metamonada, consistent with ancient inheritance, but the Mnd1 protein phylogeny did not completely reflect the expected species relationships. Dmc1 showed evidence of duplication within Fornicata, but multiple copies were also identified in other metamonad lineages. Similarly, Metamonada sequences did not form a single monophyletic group in either the Hop2 or Dmc1 phylogeny, instead occurring in separate regions of the trees that differed from the relationships recovered in previously published species phylogenies (Salas-Leiva et al., 2021). Overall, the three protein phylogenies suggest that these functionally related proteins do not necessarily share the same evolutionary history within Metamonada.

Salas-Leiva et al. (2021) previously identified Dmc1 duplication in Fornicata, and they proposed that this may have enabled a Dmc1-dependent homologous recombination mechanism in the last common ancestor of Fornicata. The Dmc1 phylogeny supports the presence of multiple Dmc1 copies within Fornicata. However, Fornicata sequences occurred in multiple separated regions of that phylogeny, and evidence of multiple Dmc1 copies was found in several other metamonad lineages. In BaSk, corresponding *Skoliomonas* sequences occurred in separate regions of the tree, and multiple *Barthelona* copies were also identified. Similarly, Preaxostyla/Oxymonadida contained separate Dmc1 clades with corresponding copies from several of the same taxa, including Pyrs and OxNc. These patterns suggest that Dmc1 duplication may have occurred in multiple metamonad lineages, although the timing and number of duplication events remain uncertain. This broader distribution was unexpected and suggests that duplication and differential retention or loss may have played a larger role in Dmc1 evolution across Metamonada than originally hypothesized.

The Hop2 and Mnd1 phylogenies appeared more similar to each other than to Dmc1, which is interesting given that Hop2 and Mnd1 form a heterodimer that promotes homologous recombination (Tsubouchi and Roeder, 2002). However, their evolutionary patterns were not identical. Mnd1 was represented across Parabasalia, Fornicata, BaSk, Anaeramoebae, and Preaxostyla, consistent with an ancient and broad distribution within Metamonada, whereas Hop2 was not identified in all metamonad groups, and the lineages it represented occurred in separate regions of the phylogeny. For example, Parabasalia was more closely related to Preaxostyla in the Hop2 phylogeny, although this relationship had weak bootstrap support. Therefore, while the apparent similarity between Hop2 and Mnd1 is consistent with a related evolutionary history, current phylogenies do not provide sufficient evidence to conclude that the two proteins co-evolved.

Several limitations prevent us from fully inferring evolutionary ancestry or explaining unusual phylogenetic placements. These include weak node support for deeper relationships outside the Metamonada clades, limited taxon and sequence sampling, and differences in rooting

among the three phylogenies. While the Hop2 phylogeny was rooted between Diphora+ and Opimoda+, the Mnd1 and Dmc1 phylogenies were not assigned biological roots. These limitations make it difficult to determine whether unusual placements reflect gene loss, duplication, divergence, or lateral gene transfer (LGT). In the future, expanded taxon and sequence sampling, particularly within BaSk and Anaeramoebae, where Hop2 representation was limited or absent, could help improve the resolution of these relationships and determine whether separated copies represent ancient duplications followed by differential loss or multiple independent duplication events. A more systematic comparison of all three protein phylogenies could determine whether patterns of duplication, loss, and divergence correspond across proteins and provide a stronger picture of the evolution of this recombination system in Metamonada. Overall, our findings suggest that although Hop2, Mnd1, and Dmc1 function together in homologous recombination, their evolutionary histories within Metamonada are complex and may not have followed a single shared evolutionary path.

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